MILITARY SPECIFICATION

CONTROL, POWER SUPPLY: 10559261

1. SCOPE

1.1 This specification covers one type of electronic unit known as Control, Power Supply: 10559261 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

F10559261	Control, Power Supply

FSC 1240

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PACKAGING DATA SHEET

10559261 Packaging of Control, Power Supply: 10559261

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The power supply control shall be manufactured in accordance with Drawing F10559261 and drawings pertaining thereto.

3.2 General specification. - The power supply control shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Part identification and marking

3.2.1 Electronic workmanship. - The power supply control shall meet the workmanship requirements of Drawing F10559261 and drawings pertaining thereto.

3.3 Performance. - Unless otherwise specified, the power supply control shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (^{o}F) and 90°F.

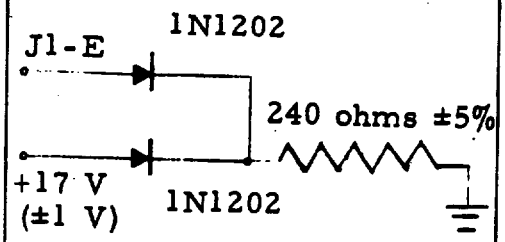
3.3.1 Loads, power and signals. - The power supply control shall perform as specified herein when the loads, power and signals of Table I are applied as specified.

TABLE I

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
1.	Loads:		As specified herein
1.1	Resistor	2.2 kilohms, ±5%	

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TABLE I (continued)

ITEM	CONDITION	CHARACTERISTICS		CONNECTIONS
1.2	Diode-resistor network			
2.	Power sources:	Tolerance	Maximum p-p ripple	Connected between following pins of J1:
2.1	15 Vdc	±0.7 Vdc	25 mV	F(+) and B(-)
2.2	29 Vdc	±1 Vdc	25 mV	G(+) and B(-)
2.3	17 Vdc	±1 Vdc	25 mV	(See item 1.2 above)
3.	Signal sources:			
3.1	Analog	±14 Vdc to +32 Vdc, adjustable		As specified herein.

3.3.1.1 Battery charging current.

3.3.1.1.1 With the signal voltage of Table I adjusted to 24 ±1 Vdc and applied between J1-H(+) and J1-B(-), and the 2.2-kilohm load of Table I connected between J1-A and J1-B, the dc current at J1-G shall be 47 ±10 mA.

3.3.1.1.2 Under the conditions of 3.3.1.1.1, a reduction of the temperature of thermostat S1 to less than 14°F shall cause the current at J1-G to decrease by at least 10 mA.

3.3.1.2 Q-switch voltage.

3.3.1.2.1 Under the conditions of 3.3.1.1.1, but with the signal voltage of Table I adjusted to 31 ±1 Vdc and applied between J1-H(+) and J1-B(-), the voltage at J1-A shall be 30.5 ±1.5 Vdc.

3.3.1.2.2 Under the conditions of 3.3.1.2.1, but with the signal voltage of Table I adjusted to 15 ±0.5 Vdc and applied between J1-H(+) and J1-B(-), the voltage at J1-A shall be 23 ±6 Vdc.

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3.3.1.3 +24-volt output.

3.3.1.3.1 With the diode-resistor load of Table I connected as specified, and with the signal voltage of Table I adjusted to 24 ±1 Vdc and applied between J1-H(+) and J1-B(-), the voltage across the 240-ohm load shall be 16 ±2 Vdc.

3.3.1.3.2 Under the conditions of 3.3.1.3.1, but with the voltage at J1-H adjusted to 15 ±0.5 Vdc, the voltages across the 240-ohm load and at pin J1-A shall be within 2 volts.

3.3.2 Environmental.

3.3.2.1 Storage temperature. - The power supply control shall meet the requirements of 3.3.1 at ambient temperature (60°F to 90°F) after having been exposed to and thermally stabilized at -80°F and +160°F.

3.3.2.2 Operating temperatures. - The power supply control shall meet the requirements of 3.3.1.1.2, 3.3.1.2 and 3.3.1.3 while exposed to and thermally stabilized at -40°F; and shall meet the requirements of 3.3.1.1.1, 3.3.1.2 and 3.3.1.3 while exposed to and thermally stabilized at +125°F, subsequent to which it shall meet the requirements of 3.3.1.1, 3.3.1.2, and 3.3.1.3 at ambient temperature (60°F to 90°F).

3.3.2.3 Vibration. - The power supply control shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

(a)	Amplitude:	0.5 inch maximum
(b)	Acceleration:	4g maximum
(c)	Crossover frequency:	12.5 Hertz (Hz)
(d)	Sweep time:	5 to 500 to 5 Hz in 15 minutes. Two sweeps in each of the three mutually perpendicular axes.

3.3.2.4 Shock. - The power supply control shall meet the requirements of 3.3.1 after being exposed to a total of 6 shock impulses, one in each direction of the three mutually perpendicular axes. Each shock impulse shall be a sawtooth or half-sine wave with a time duration of 6 milliseconds. The peak amplitude of each shock impulse shall be 100g.

3.3.2.5 Immersion. - With the power supply control pressurized to 5 ±1 psig with nitrogen or air, and submerged no more than 2 feet in water at a temperature of 110 ±10°F, there shall be no evidence of gas bubbles emanating from inside the unit during a 15-minute observation period.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three power supply controls. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II, III, and IV herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III, and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. - The requirements and tests in Table II shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE II

(100 percent inspection)

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Battery charging current	3.3.1.1	4.6.2
102.	Q-switch voltage	3.3.1.2	4.6.3
103.	+24-volt output	3.3.1.3	4.6.4

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. - One power supply control shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table III.

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TABLE III

	CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental. - Three power supply controls shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table IV.

TABLE IV

	CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature	3.3.2.1	4.6.5.1
304.	Operating temperature	3.3.2.2	4.6.5.1
305.	Vibration	3.3.2.3	4.6.5.2
306.	Shock	3.3.2.4	4.6.5.3
307.	Immersion	3.3.2.5	4.6.5.4

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the technical data package (TDP).

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4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment. - In conjunction with 4.5.2, the standard test equipment listed in Table V shall be utilized in performing the applicable test of 4.6.

TABLE V

	NOMENCLATURE	DESCRIPTION
1.	D-C power supply	Plus 15 ±0.7 volts Ripple: 25 mV p-p max
2.	D-C power supply	Plus 29 ±1 volts Ripple: 25 mV p-p max
3.	D-C power supply	Plus 17 ±1 Vdc Ripple: 25 mV p-p max
4.	D-C ammeter	Range: 0 to 100 mA Accuracy: ±20%
5.	D-C voltmeter	Range: 0 to 50 volts Accuracy: ±0.3%
6.	Signal source	Capable of providing the dc level of amplitude specified in Table I (item 3.1).

4.6 Test methods and procedures.

4.6.1 Test conditions. - Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

4.6.1.1 Initial conditions. - Connect the power to the power supply control as specified in Table I.

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4.6.2 Battery charging current.

4.6.2.1 Connect the 2.2-kilohm load of Table I between J1-A and J1-B, and apply the signal voltage of Table I, adjusted to 24 ±1 Vdc, between J1-H(+) and J1-B(-). The dc current at J1-G shall meet the requirements of 3.3.1.1.1.

4.6.2.2 Under the conditions of 4.6.2.1, reduce the temperature of thermostat S1 to less than 14°F. The dc current of J1-G shall meet the requirements of 3.3.1.1.2.

4.6.3 Q-switch voltage.

4.6.3.1 Under the conditions of 4.6.2.1, adjust the signal voltage of Table I to 31 ±1 Vdc. The voltage at J1-A shall meet the requirements of 3.3.1.2.1.

4.6.3.2 Under the conditions of 4.6.3.1, adjust the signal voltage of Table I to 15 ±0.5 Vdc. The voltage at J1-A shall meet the requirements of 3.3.1.2.2.

4.6.4 +24-volt output.

4.6.4.1 Connect the diode-resistor load of Table I as specified, and apply the signal voltage of Table I, adjusted to 24 ±1 Vdc, between J1-H(+) and J1-B(-). The voltage across the 240-ohm load shall meet the requirements of 3.3.1.3.2.

4.6.4.2 Under the conditions of 4.6.4.1, adjust the voltage at J1-H to 15 ±0.5 Vdc. The voltage across the 240-ohm load shall meet the requirements of 3.3.1.3.2.

4.6.5 Environmental tests. - The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 except that the power supply control may be mounted on an inside surface of the test chamber for the temperature test and the tolerance requirements for vibration frequency measuring instruments shall not apply.

4.6.5.1 Temperature test. -

NOTE: The rate of temperature change shall be less than 7°F per minute during temperature cycling.

(a) Place the power supply control in the test chamber and reduce the chamber temperature to -80°F.
(b) Soak the power supply control at this temperature for at least 5 hours with the unit not operating.

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(c) Raise the chamber temperature to -40°F.
(d) Following thermal stabilization at -40°F, place the power supply control in operation and perform the tests specified in 4.6.2.2, 4.6.3, and 4.6.4.
(e) Raise the chamber temperature to +160°F.
(f) Soak the power supply control at +160°F for at least 5 hours with the assembly not operating.
(g) Reduce the chamber temperature to +125°F.
(h) Following thermal stabilization at +125°F, place the power supply control in operation and perform the tests specified in 4.6.2.1, 4.6.3, and 4.6.4.
(i) Reduce the chamber temperature to ambient.
(j) Following stabilization at ambient temperature, place the power supply control in operation and perform the tests specified in 4.6.2, 4.6.3 and 4.6.4.

The power supply control shall meet the requirements of 3.3.2.1 and 3.3.2.2.

4.6.5.2 Vibration. - With the power supply control firmly mounted to the vibration table perform the following:

(a) Test in accordance with Method 514-I of MIL-STD-810 for category (f) equipment, except that the acceleration shall not exceed ±4g and the maximum excursion (double amplitude) shall be 0.5 inch.
(b) Sweep the frequency from 5 Hz to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.
(c) Following vibration cycling, perform a visual-tactile examination for damage and loose parts.
(d) Place the power supply control in operation and perform the tests specified in 4.6.2, 4.6.3 and 4.6.4.

The power supply control shall meet the requirements of 3.3.2.3.

4.6.5.3 Shock. - With the power supply control rigidly mounted to the shock machine table, perform the following:

(a) Test in accordance with Method 516, Procedure IV, of MIL-STD-810, except that the power supply control shall be subjected to one shock in each direction of the three mutually perpendicular axes and the shock impulse shall be a sawtooth or half-sine wave with a duration of 6 milliseconds and a peak amplitude of 100g.

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(b) After exposure to shock, perform a visual-tactile examination for damage and loose parts.
(c) Place the power supply control in operation and perform the tests specified in 4.6.2, 4.6.3 and 4.6.4.

The power supply control shall meet the requirements of 3.3.2.4.

4.6.5.4 Immersion. - Connect the leak test port of the power supply control to a cylinder of nitrogen or air and pressurize it to 5 ±1 psig. Submerge the unit in water at 110 ±10°F to a depth not to exceed 2 feet and observe the unit for evidence of leakage (gas bubbles) for a period of 15 minutes. (Note: The unit shall not be left unattended during immersion.) The unit shall meet the leakage requirements of 3.3.2.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10559261. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The power supply control is a unit of the Laser Range Finder AN/VVS-1. It contains the 24-volt battery and associated control circuitry which provides auxiliary power to maintain laser system operation when the prime power source falls below the 18-volt minimum required for system operation. During normal operating conditions the battery is continuously charged.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet number (see 5.1).
(c) Selection of applicable levels of preservation, packaging and packing.
(d) Applicable stock number.
(e) Requirement for First Article Submission.

Custodian:

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Preparing activity

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Project No. 1240-A135

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